SCSI 1113: COMPUTATIONAL MATHEMATICS

~~**20172018-2**~~

~~**TUTORIAL 5**~~

~~**Due date: 24th May 2018**~~

Sem 1 Sesi 2021/2022
Assignment 7 Differentation and Integration

QUESTION 1 **(15 marks)**

Given the data in Table 1

Table 1

x	-2	-1.5	-1	-0.5	0	.5	1	1.5	2
y	-31	-24.4	-20	-17.1	-15	-12.9	-10	-5.6	1

a) Compute the first derivative using forward different 5 point formula x = -1 for h = 0.5 (3 marks)

b) Determine the second derivative at x = -0.5 for h = 1.0 using 3 point central different formula. (3 marks)

c) Determine the second derivative at x = -0.5 for h =0.5 using 5 point central different formula. (3 marks)

d) Determine the first derivative for $y = x^3 + 4x - 15$ and compare the result with an estimation obtained in (a). (3 marks)

e) Determine the second derivative for $y = x^3 + 4x - 15$ and compare the result with an estimation obtained in (b) and (c). (3 marks)

QUESTION 2 **(10 marks)**

Given $f(x) = \dfrac{x+2}{x-10}$

a) Approximate $f'(0.5)$ by using three-point central difference formula with h = 0.1. (5 marks)

b) Approximate $f''(0.5)$ by using three point central difference formula with h = 0.1. (5 marks)

Do calculations in 4 decimal points.

QUESTION 3 **(15 marks)**

Coordinate of points along a curve is given in Table 2 below.

x	0.0	0.2	0.4	0.6	0.8	1.0	1.2
$f(x)$	1.0	1.2	1.4	1.6	1.0	1.7	1.3

Calculate $\int_{0.0}^{1.2} f(x)dx$ using

a) Trapezoidal rule b) Simpson 1/3 rule c) Simpson 3/8 rule

QUESTION 4 **(10 marks)**

Consider $\int_1^2 \frac{1}{x}dx$. Do the calculation using Romberg method's till $R_{3,3}$ or $\varepsilon = 0.00001$. Find the error if the exact value for $\int_1^2 \frac{1}{x}dx$ is *ln* 2. Do calculations in 5 decimal points.

QUESTION 5 **(5 marks)**

Approximate $\int_1^3 x^{\frac{3}{2}}dx$ using Romberg Intergration, $R_{3,3}$. Compare the approximate result with the exact value. Do calculation in 4 decimal points. Given $R_{2,2} = 5.8354$ and $R_{3,2} = 5.8354$.

QUESTION 6 **(10 marks)**

Integrate

$$f(x) = 0.2 + 25x + 3x^2 + 2x^4$$

from $a = 0$ to $b = 2$, given $n = 8$ by using

a) Trapezoidal approximation. (4 marks)
b) Simpson's 1/3 approximation (4 marks)
c) Compare the approximate solutions in (i) and (ii) with the exact solution. Which one is better, Trapezoidal approximation or Simpson's 1/3 approximation? (2 marks)